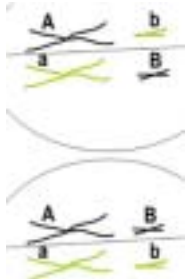


6(a)(i)	AAb^hb^h Aab^hb^h AAb^hb Aab^hb ; ;	2
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Question	Answer	Marks																									
6(a)(ii)	long, white / albino ;	1																									
6(b)	1 F1 parent genotypes: AaBb x AaBb ; 2 gametes: A B A b a B a b A B A b a B a b 3 F2 offspring genotypes:<table><tr><td></td><td>AB</td><td>Ab</td><td>aB</td><td>ab</td></tr><tr><td>AB</td><td>AABB</td><td>AABb</td><td>AaBB</td><td>AaBb</td></tr><tr><td>Ab</td><td>AABb</td><td>AAbb</td><td>AaBb</td><td>Aabb</td></tr><tr><td>aB</td><td>AaBB</td><td>AaBb</td><td>aaBB</td><td>aaBb</td></tr><tr><td>ab</td><td>AaBb</td><td>Aabb</td><td>aaBb</td><td>aabb ;</td></tr></table> 4 evidence of deduction of phenotypes from (min 9 different) genotypes ; 5 F2 offspring phenotype ratio: 9 black short : 3 white short : 3 black long : 1 white long : 5		AB	Ab	aB	ab	AB	AABB	AABb	AaBB	AaBb	Ab	AABb	AAbb	AaBb	Aabb	aB	AaBB	AaBb	aaBB	aaBb	ab	AaBb	Aabb	aaBb	aabb ;	5
	AB	Ab	aB	ab																							
AB	AABB	AABb	AaBB	AaBb																							
Ab	AABb	AAbb	AaBb	Aabb																							
aB	AaBB	AaBb	aaBB	aaBb																							
ab	AaBb	Aabb	aaBb	aabb ;																							

Question	Answer	Marks																									
6(c)(i)	any three from: 1 because they are, on separate chromosomes / not linked ; 2 each, bivalent / <u>pair</u> of homologous chromosomes, orients itself separately ; 3 (at equator) in metaphase I of meiosis ; 4 gives, four / different / new, combinations of alleles / gametes ; 5 parental allele combinations are not, preserved / fixed ; 6 AVP ; 	3																									
6(c)(ii)	<table><tr><th>phenotype</th><th>observed number (O)</th><th>expected number (E)</th><th>$(O - E)^2$</th><th>$\frac{(O - E)^2}{E}$</th></tr><tr><td>short, black fur</td><td>25</td><td>27</td><td>4</td><td>0.148</td></tr><tr><td>long, black fur</td><td>11</td><td>9</td><td>4</td><td>0.444</td></tr><tr><td>short, white fur</td><td>8</td><td>9</td><td>1</td><td>0.111</td></tr><tr><td>long, white long</td><td>4</td><td>3 ;</td><td>1</td><td>0.333</td></tr></table>	phenotype	observed number (O)	expected number (E)	$(O - E)^2$	$\frac{(O - E)^2}{E}$	short, black fur	25	27	4	0.148	long, black fur	11	9	4	0.444	short, white fur	8	9	1	0.111	long, white long	4	3 ;	1	0.333	1
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6(c)(iii)	1.036 ;	1																									